Reinforcement Learning

Think of the most mundane thing that you probably do in your life, say making butter toast. If you explore this simple activity you will learn that there is a deep hierarchy of learned behaviours that determines how you perform this task.

On one level you have the act of making toast. How much butter do you add, how long do you toast it on a stove? How do you know its done? Part of this is based on what you've learned from others, but most important is what comes from personal taste. You've made toast before, and you know what you like. Each successfully toasted piece of bread counts as positive reinforcement, and strengthens your preference for the amount of butter you used and the colour of the bread after toasting.

On a lower level, you learned how to make a bread toast. The fact that it involves butter and bread, that you have to turn on the gas and put on a frying pan. On an even lower level you probably had to learn that you need to go the kitchen to do all this, and how to turn on the gas in the first place. Go lower and you come to learning how to move muscles to make movements such as walking or turning knobs.

Computer programs, especially those used in artificial intelligence can also be developed with this approach in mind. With reinforcement learning the computer performs a task and gets a positive or negative response which it then uses to categorise that action appropriately. For example a AI chess program could learn whether a move is good or bad based on whether that move leads to a victory or defeat.

This poses some interesting problems though. Firstly, which move is it that the computer made that resulted in victory? It's possible that the move that led to victory came dozens of moves before the final checkmate. Where should it assign credit? Secondly, is it better to try something new, and explore new moves or just use old winning moves even though they might be countered this time?

Conclusion

Machine Learning is in use more places than you probably realised, and it is a rich subject for study. There is yet much to be discovered

Here is an example of optical character recognition on a number plate. Before you can begin recognising characters, you first have to break the image into individual characters. This too can be accomplished with machine learning.

in this field especially when it comes to things such as reinforcement learning.

There are dozens if not hundreds of algorithms for different kinds of machine learning. Such as Neural Networks, Support Vector Machines, k-means clustering, Linear and Logistic regression, etc.

There are also many software packages that help you apply these algorithms to data to make sense of it. Examples of these would be scikit-learn, OpenCV, MLPACK, Apache Mahout, IBM SPSS, and MATLAB among hundreds of others.

If anything there is too much data available these days, and Machine Learning is the way to make sense of it and process it when it reaches the point when humans no longer can.

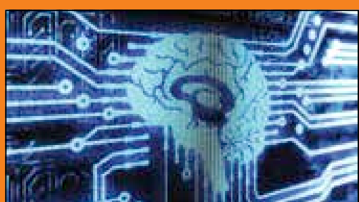
If you really want to learn machine learning in depth, there are excellent free resources available online. There are free courses on Machine Learning available on educational websites such as edx.org and coursera.org that are well worth checking out. >

*Training Data

*Coursera's Machine Learning Course

>>Coursera has a great course on machine learning that uses MATLAB and is easy for beginners to get into. It includes topics such as Perceptrons, Neural Networks, and Support Vector Machines.

http://dgit.in/MachineLearn



*EDX's Machine Learning Course

>>This course on Machine Learning at EdX focuses on the theoretical aspects of Machine Learning, what learning even means and if it is possible for a machine to learn. It is available free of cost.

http://dgit.in/EDXMachLearn

*Sage

>>Sage is a great software package for mathematics and science. It tries to rival MATLAB is features, while using Python as its language instead. It comes with a plethora of scientific packages for Python and you can add more.

http://www.sagemath.org/